Class 10 : Structural Bioinformatics (pt.1)

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The PDB database

The Protein Data Bank (PDB) is the main repository of biomolecular structure data. Let's see what's in it:

```
stats <- read.csv("pdbstats.csv", row.names = 1)
stats
```

```
                        X.ray    EM   NMR Integrative Multiple.methods Neutron
Protein (only)         178795 21825 12773         343               226      84
Protein/Oligosaccharide  10363  3564    34           8                11       1
Protein/NA                9106  6335   287          24                 7       0
Nucleic acid (only)       3132   221  1566           3                15       3
Other                      175    25    33           4                 0       0
Oligosaccharide (only)      11     0     6           0                 1       0
                       Other  Total
Protein (only)            32 214078
Protein/Oligosaccharide    0  13981
Protein/NA                 0  15759
Nucleic acid (only)        1   4941
Other                      0    237
Oligosaccharide (only)     4     22
```

```r
n.sums <- colSums(stats)
n.sums/n.sums["Total"]
```

```
           X.ray               EM              NMR      Integrative
    0.8095077464     0.1283842935     0.0590278614     0.0015340257
Multiple.methods          Neutron            Other            Total
    0.0010441012     0.0003533881     0.0001485836     1.0000000000
```

```r
round(n.sums,digits=2)
```

```
           X.ray               EM              NMR      Integrative
          201582            31970            14699              382
Multiple.methods          Neutron            Other            Total
             260               88               37           249018
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```r
100 * n.sums[c("X.ray", "EM")] / n.sums["Total"]
```

```
   X.ray       EM
80.95077 12.83843
```

Q2: What proportion of structures in the PDB are protein?

```r
stats <- read.csv("pdbstats.csv", row.names = 1)
total_structures <- sum(stats$Total)
protein_rows <- grep("^Protein", rownames(stats))
protein_total <- sum(stats[protein_rows, "Total"])
protein_total / total_structures
```

```
[1] 0.979118
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

The query matched 5 structures.

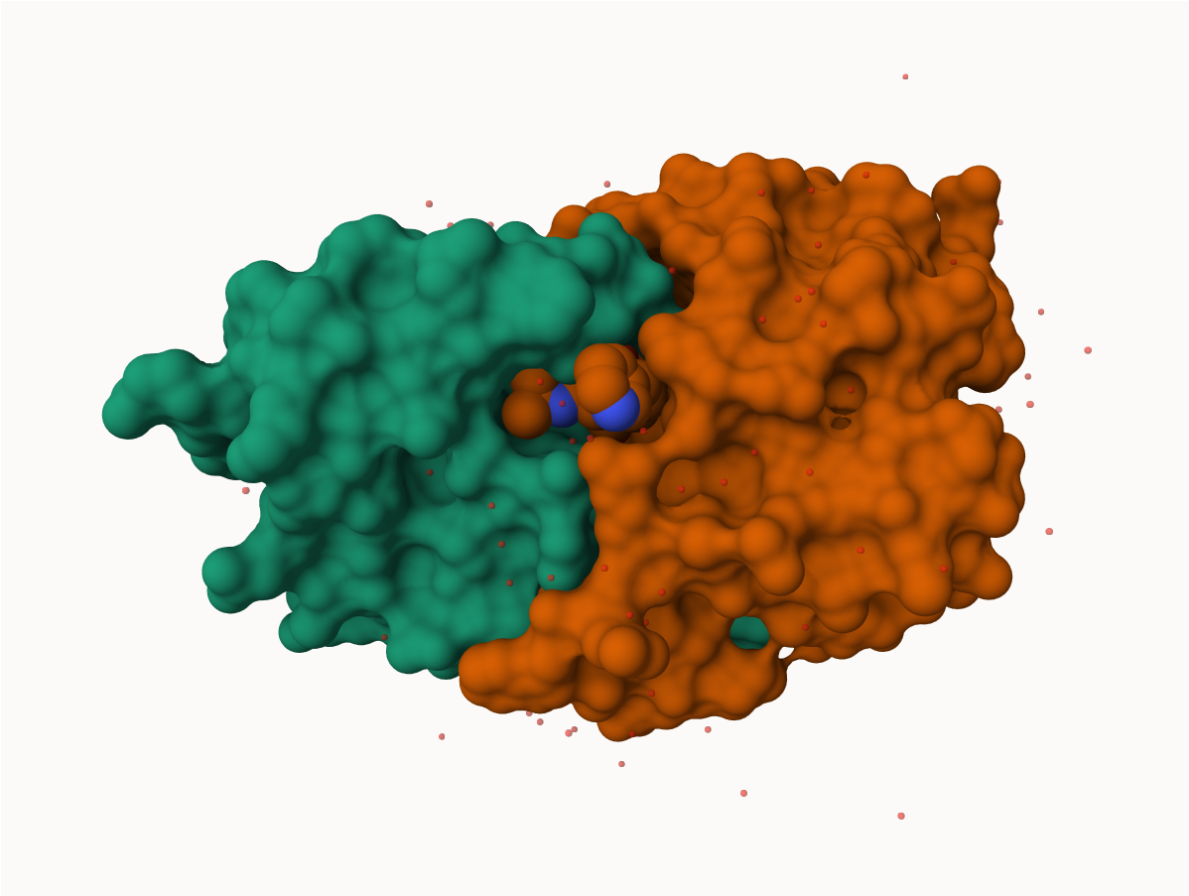
Q. What is the total number of entries?

```
n.sums["Total"]
```

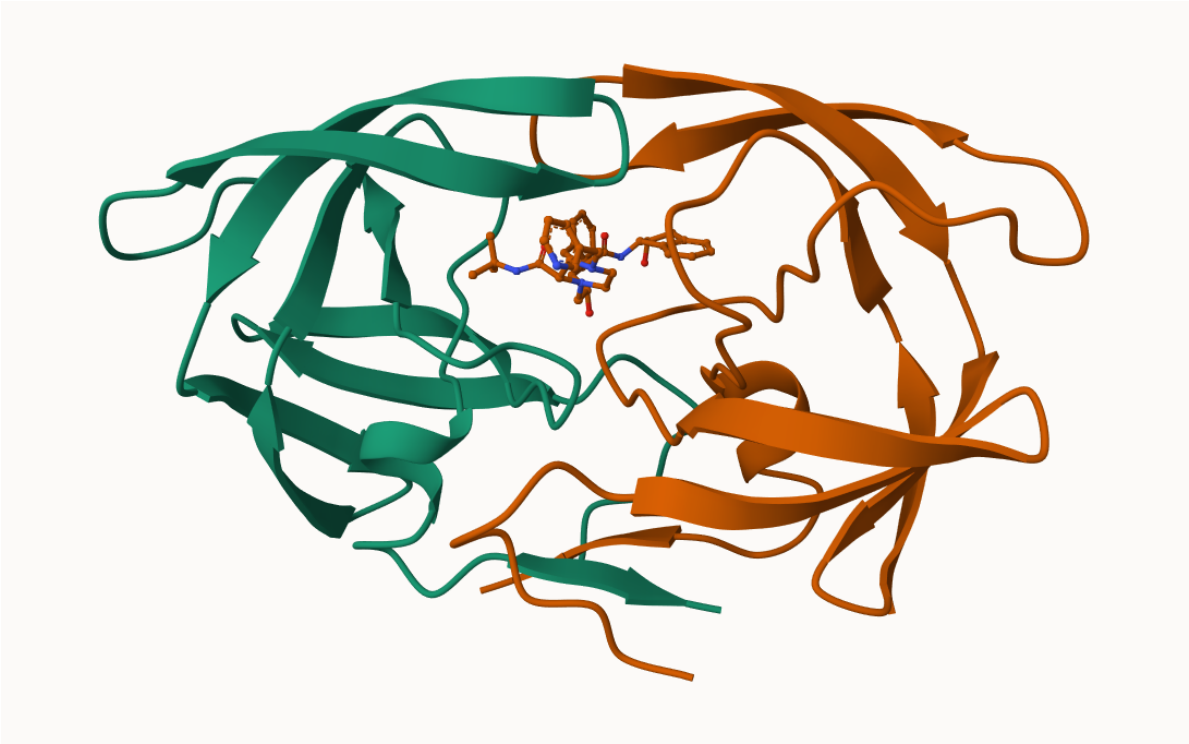
```
 Total
249018
```

Using Molstar

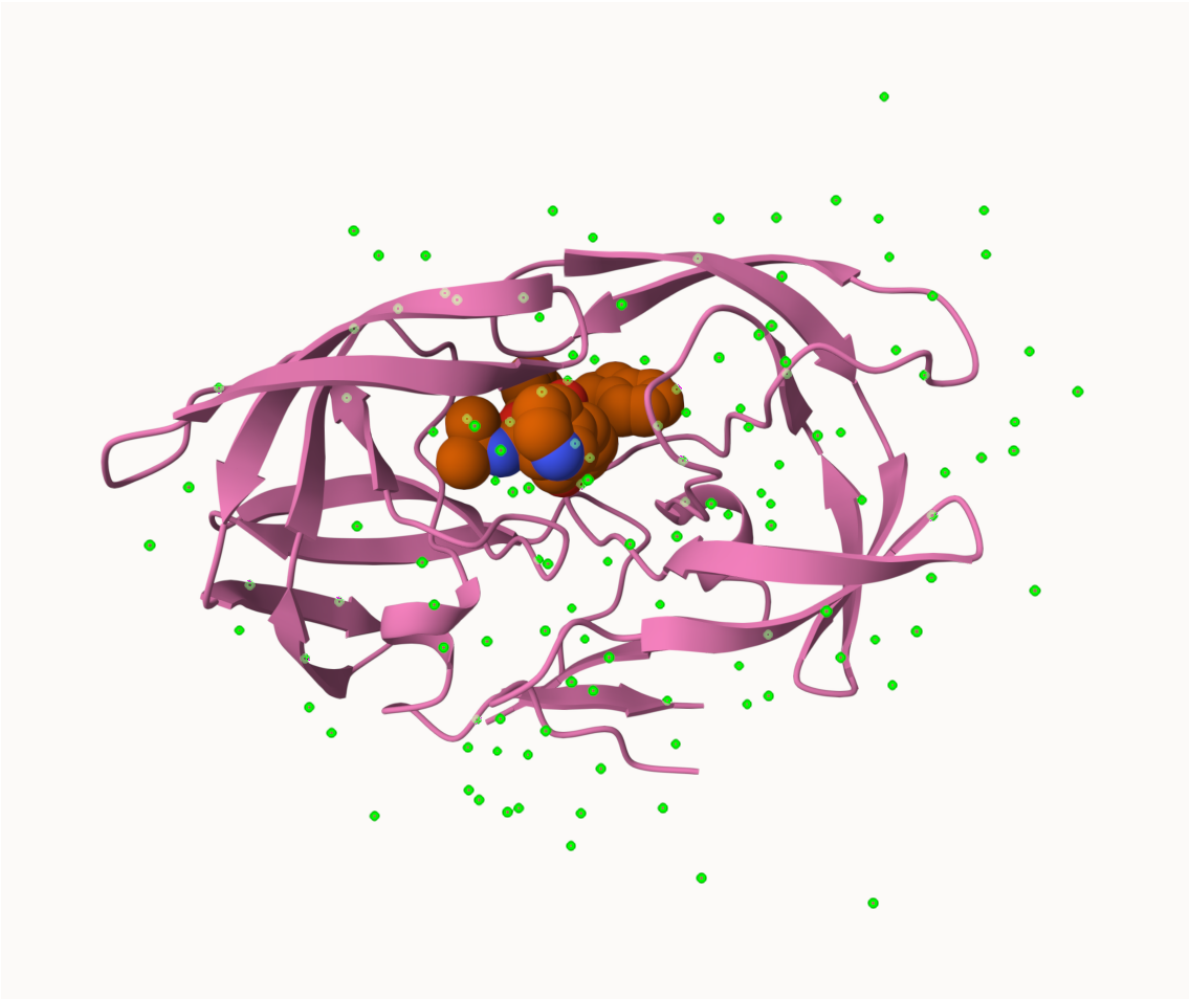
We can use the main Molstar viewer online:



> Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing the inhibitor (ligand) in ball and stick.



Q. One final image showing catalytic APS 25 and all-important active site water molecule as spacefill.



Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

In PDB visualization h2o appears as a single atom.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

HOH 127

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document. Discussion Topic:

> Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

HIV protease has flexible flaps covering the active site. These flaps can open temporarily, allowing substrates and inhibitors to slide into the catalytic pocket. Larger ligands may require flap displacement, which is why dynamics are important in inhibitor design.

> Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

HIV protease is a homodimer, so certain beta strands and loops form only at the dimer interface. Dimer only elements would not be stable in a monomer.

Intro to Bio3d

```r
library(bio3d)

hiv <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
hiv
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
```

```
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(hiv$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
pdbseq(hiv)
```

```
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K"
 21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G"
 41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D"
 61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
 81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99   1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
  2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
 22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40  41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
 42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
```

```
 62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
 82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH and MK1

Q9: How many protein chains are in this structure?

2 chains, A and B

Let's try out the new **bio3dview** package that is not yet on CRAN. We'll use the **remotes** package to install any R package from GitHub.

Quick viewing of PDBs

```
library(bio3dview)
sele <- atom.select(hiv, resno=25)
#view.pdb(hiv, backgroundColor ="pink",
         #highlight=sele,
        # highlight.style = "spacefill")
```

Prediction of Protein flexibility

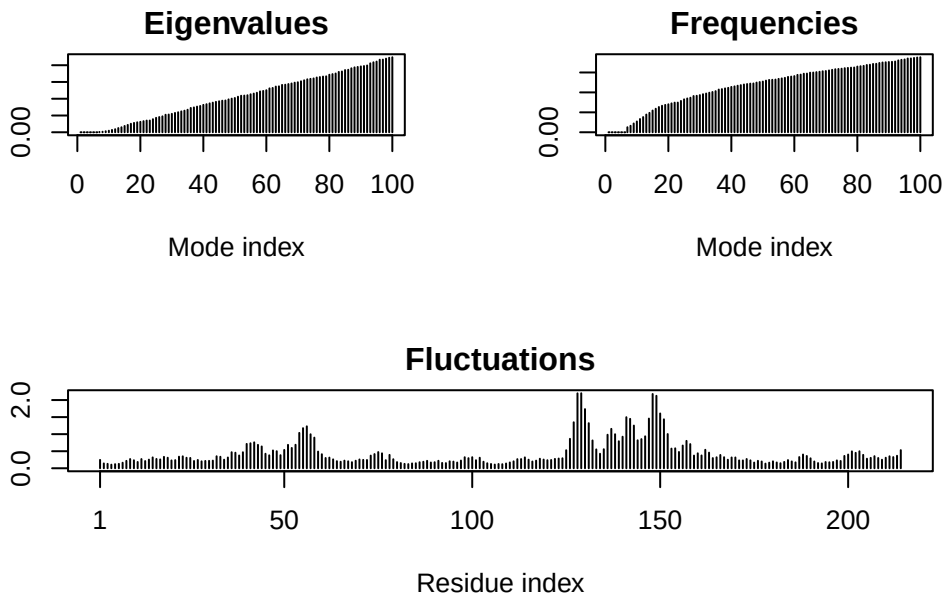
```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.008 seconds.
 Diagonalizing Hessian...   Done in 0.03 seconds.
```

```r
plot(m)
```



Write out our results as a small trajectory movie:

```r
mktrj(m, file="results.pdb")
```

```r
#view.nma(m, pdb=adk)
```